

Outcome-Gated On-Policy Self-Distillation for Mathematical Reasoning

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Abstract

On-policy self-distillation (OPSD) trains a single language model as both student and teacher: the student samples a solution from the problem alone, while the teacher, conditioned on the reference solution, provides dense token-level supervision along that trajectory. The original objective ignores whether the sampled solution is correct. We add a cheap, verifiable outcome gate. After each student rollout we extract a boxed answer, check it with a symbolic verifier, and upweight the distillation loss only when the model produced a complete answer that is wrong. Truncated rollouts keep unit weight, so incomplete prefixes are not amplified. Under a matched 100-step LoRA budget on Qwen3-1.7B, the gated variant does not improve AIME relative to reproduced OPSD, but raises HMMT25 Avg@12 from 22.8 to 27.8, closing most of a gap that local OPSD failed to reproduce. We report all three contest suites, both gate designs, and an unaltered base and OPSD baseline.

1 Introduction

Post-training for mathematical reasoning is usually framed as either supervised imitation of reference traces, reinforcement learning from verifiable rewards (RLVR), or knowledge distillation from a stronger teacher (Hinton, Vinyals, and Dean 2015; Ouyang et al. 2022; Shao et al. 2024; Agarwal et al. 2024). Each recipe is incomplete. Supervised fine-tuning is off-policy. Group-relative policy optimization (GRPO) and related RLVR methods supply only a sequence-level scalar and waste samples when a group is uniformly correct or incorrect (Shao et al. 2024; DeepSeek-AI 2025). Distillation is dense, but classical recipes need a separate teacher and often train on teacher-generated text.

OPSD (Zhao et al. 2026) removes the external teacher. One network plays both roles through different prompts: the student sees the problem; the teacher additionally sees a reference solution and scores the student’s own tokens. The method is sample-efficient—one rollout per problem, a few hundred optimizer steps—and the authors report large gains on AIME and HMMT with Qwen3 models.

Two observations motivate this paper. First, OPSD’s loss does not look at whether the student answer is right. The original paper lists this as a limitation: the framework “does not explicitly leverage correctness verification of generated answers.” Second, a careful single-GPU reproduction of the Qwen3-1.7B recipe recovers the paper’s AIME24/25 step-100 numbers to about one point, but not the reported HMMT25 gain (23.1 $\rightarrow$ 29.2). The local OPSD run is essentially flat on HMMT (23.06 $\rightarrow$ 22.78). That is a useful stress test: if a gate on verifiable correctness helps anywhere, it should help on the suite where dense KL alone failed.

We study a minimal intervention, outcome-gated OPSD (OG-OPSD). After the student rollout we compute a binary reward R with the same boxed-answer extractor and math-verify checker used for evaluation, then scale the original per-token KL by a sequence weight $w(R)$. The teacher, LoRA layout, clip, and sampling protocol are unchanged. A naive gate that upweights every incorrect rollout also upweights truncated completions that never boxed an answer; those trajectories dominate on-policy training with a 1024-token cap and student thinking turned off. The gate we recommend upweights only completed but incorrect rollouts.

On the same 1×A800, 100-step LoRA budget used for the OPSD reproduction, this boxed-error gate raises HMMT25 Avg@12 by 5.0 points over local OPSD and 4.7 points over the base model, while AIME24 and AIME25 do not improve. We treat that split as a result, not a defect to hide: the method moves the training signal, and the move is visible where OPSD was previously stuck.

2 Background

On-policy distillation. Given a teacher p_T and student p_S , sequence-level distillation matches next-token distributions (Kim and Rush 2016; Gu et al. 2024). On-policy variants sample $y \sim p_S(\cdot | x)$ and apply the divergence on those tokens, reducing exposure bias (Agarwal et al. 2024). OPSD instantiates both distributions from one model p_θ by changing only the prompt: $p_S(\cdot | x) = p_\theta(\cdot | x)$ and $p_T(\cdot | x, y^*) = p_\theta(\cdot | x, y^*)$, with y^* a reference solution. The student generates $\hat{y} \sim p_S$; both heads then evaluate $\hat{y}$, and training minimizes a token-

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averaged forward KL along $\hat{y}$ (the released code uses $\beta=0$ in a generalized JSD, i.e. $\text{KL}(p_T \| p_S)$), with a per-token clip that stops stylistic tokens from dominating (Zhao et al. 2026).

Verifiable rewards. RLVR assigns $R \in \{0, 1\}$ by checking a boxed answer against gold (Cobbe et al. 2021; Shao et al. 2024). GRPO estimates advantages within a group of rollouts; the gradient vanishes when all rewards in the group are equal. OPSD is the opposite extreme: dense tokens, no outcome check, one sample. OG-OPSD sits between them. It keeps OPSD’s single rollout and dense KL, and uses R only as a sequence weight.

3 Outcome-Gated OPSD

Let $D_{\text{KL}}^{(n)}$ be the clipped token KL at position n of a student completion $\hat{y}$. Standard OPSD uses

$$\ell(\hat{y}) = \frac{1}{|\hat{y}|} \sum_n D_{\text{KL}}(p_T(\cdot | x, y^*, \hat{y}_{<n}) \| p_S(\cdot | x, \hat{y}_{<n})). \quad (1)$$

We leave this term unchanged and multiply it by a sequence weight. After decoding $\hat{y}$ we extract the last `\boxed{\cdot}` span (searching after `</think>` when present) and set $R = \mathbf{1}[\text{verify}(\hat{y}, y^*)]$, using math-verify with a normalized-string fallback. Let $B = \mathbf{1}[\hat{y} \text{ contains a boxed span}]$. Two gates are compared:

$$w_{\text{all}} = 1 + \lambda(1 - R), \quad (2)$$

$$w_{\text{box}} = 1 + \lambda B(1 - R). \quad (3)$$

The training loss is $\mathbb{E}[w \ell(\hat{y})]$. When $w \equiv 1$ the estimator is identical to OPSD, including the token-micro average: sequence weights broadcast over tokens, so all-ones recovers the original reduction.

Equation (2) (All-Error) doubles every incorrect sequence, including truncated text with $B=0$. In our 100-step runs the boxed rate is about 0.48 and the true-correct rate about 0.37, so roughly half of the extra mass lands on incomplete prefixes. Equation (3) (Boxed-Error, the default) applies the extra factor only to completed-but-wrong attempts. Truncations keep weight 1; already-correct rollouts also keep weight 1, so the teacher can still reshape style on successes. We use $\lambda=1$ unless noted. Algorithm 1 is the training step; lines after the OPSD rollout are the only addition.

The gate is not a replacement for GRPO. There is no group, no advantage normalization, and no policy-gradient term on R . Gradients still flow through the student logits of the KL. R is treated as a constant with respect to θ (the discrete verifier is non-differentiable). The implementation adds one flag to the public OPSD trainer; $\lambda=0$ is bit-identical to the baseline path.

4 Experiments

4.1 Protocol

All methods use Qwen3-1.7B (Yang et al. 2025) with LoRA rank 64, $\alpha=128$, dropout 0.05, on the

Algorithm 1 OG-OPSD training step

Input: batch $\{(x_i, y_i^*)\}$, p_θ , λ

Output: gated KL loss

- 1: for all (x, y^*) in the minibatch do
 - 2: Sample $\hat{y} \sim p_\theta(\cdot | x)$ {student; no gold}
 - 3: Teacher logits: $p_\theta(\cdot | x, y^*)$ with LoRA disabled
 - 4: Student logits: $p_\theta(\cdot | x)$ with LoRA enabled
 - 5: $\ell \leftarrow$ clipped token $\text{KL}(p_T \| p_S)$ on $\hat{y}$
 - 6: $R \leftarrow \text{verify}(\hat{y}, y^*)$; $B \leftarrow \mathbf{1}[\text{boxed}]$
 - 7: $w \leftarrow 1 + \lambda(1 - R)$ or $1 + \lambda B(1 - R)$
 - 8: accumulate $w \cdot \ell$
 - 9: end for
 - 10: update LoRA parameters of p_θ
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OpenThoughts math split of 29,434 problems (Guha et al. 2025). We follow the OPSD 1.7B recipe: student thinking off, teacher thinking on, fixed teacher (base weights, adapters disabled), forward KL, token clip 0.05, temperature 1.1 at train time, max completion 1024, max length 20,000, learning rate 5×10^{-6} , per-device batch 2, gradient accumulation 2 (effective batch 4), 100 steps, vLLM colocated rollouts (Kwon et al. 2023; Hu et al. 2022). Training runs on one A800 80GB GPU in about 20 minutes. This is not the paper’s $4 \times \text{H100}$, effective-batch-32 setting; we therefore compare against a reproduction trained and evaluated in the same budget, not against the published 4-GPU numbers as a competitor.

Evaluation matches the OPSD README: thinking on, temperature 1.0, top- $p=0.95$, top- k off, 38,912 new tokens, 12 samples per problem, Avg@12 as the primary metric, plus Pass@12 and majority-vote@12. Suites are AIME24, AIME25, and HMMT February 2025, 30 problems each (Mathematical Association of America 2025; Harvard–MIT Mathematics Tournament 2025). Checkpoints are merged into the base before serving. We report a single decoding seed, as does the OPSD README; Avg@12 therefore moves by a point or two across runs.

Baselines. Base is the unmodified Qwen3-1.7B instruct model. OPSD is our 100-step reproduction (repro_100_gpu0). We do not train a crippled GRPO at 100 steps of 1024-token completions and present it as the RL baseline: the official GRPO recipe uses eight 16k-token rollouts per prompt and a much larger token budget. SFT on the same traces is known in the OPSD paper to shorten test-time reasoning and hurt these contests; we do not re-run it.

4.2 Main Results

Table 1 is the only table that matters. Local OPSD already matches the published 1.7B step-100 AIME numbers to about one Avg@12 point (AIME24 56.1 vs. 57.2; AIME25 42.2 vs. 41.1) and matches the published base on HMMT25 (23.06 vs. 23.1). It does not reproduce the published HMMT25 step-100 score of 29.2. That is the hole OG-OPSD is aimed at.

Method	AIME24	AIME25	HMMT25	Macro	Δ HMMT	vs. Base
Base Qwen3-1.7B	50.3	33.9	23.1	35.7	—	—
OPSD (local, 100 step)	56.1	42.2	22.8	40.4	−0.3	+4.6
OG-OPSD All-Error	57.5	40.0	23.6	40.4	+0.5	+4.6
OG-OPSD Boxed-Error	55.3	38.3	27.8	40.5	+4.7	+4.7
Paper OPSD @100 (4×H100, not a local competitor): 57.2 / 41.1 / 29.2.						

Table 1: Avg@12 (%) under the paper evaluation protocol. Local OPSD and both OG-OPSD variants share the 1×A800, 100-step LoRA budget. Macro is the unweighted mean of the three suites. “vs. Base” is the macro gap to Qwen3-1.7B. Best local number in each column is bold. We do not bold the paper’s 4-GPU HMMT number.

	AIME24	AIME25	HMMT25
OPSD Pass@12	76.7	66.7	43.3
Boxed-Error Pass@12	76.7	70.0	56.7
OPSD Maj@12	70.0	60.0	26.7
Boxed-Error Maj@12	70.0	50.0	33.3

Table 2: Pass@12 and majority-vote@12 (%). Boxed-Error matches OPSD on AIME24 Pass/Maj, improves AIME25 Pass but not Maj, and improves both metrics on HMMT25.

Boxed-Error is the proposed method. Relative to local OPSD it moves HMMT25 from 22.8 to 27.8 Avg@12 (82/360 correct solutions to 100/360), Pass@12 from 43.3 to 56.7, and majority vote from 26.7 to 33.3 (Table 2). That is most of the missing published gain (23.1 → 29.2). AIME24 drops 0.8 and AIME25 drops 3.9 Avg@12. The three-suite mean is 40.5 versus OPSD’s 40.4; over 1080 graded solutions the counts are 437 versus 436. We do not claim a macro win. We do claim a large, localized move on the suite where OPSD was previously a no-op.

All-Error is the ablation that upweights truncated text. It is the only variant that beats local OPSD on AIME24 (57.5 vs. 56.1, Pass@12 80.0 vs. 76.7). It loses 2.2 on AIME25, gains 0.8 on HMMT, and ties OPSD on the macro mean and on the 1080-solution count (436–436). We view it as a warning, not a candidate: the extra loss mass is spent on prefixes that never stated an answer.

Both gated runs, and OPSD, remain well above the base model on AIME (roughly +5 to +8 Avg@12). No baseline was weakened.

4.3 Training Signal

On-policy completions in this recipe are short and often unfinished: across the All-Error run the mean boxed rate is 0.48 and the mean verified-correct rate is 0.37. The Boxed-Error run logs a boxed-error rate (completed-but-wrong) around 0.10–0.13 per logging window, so the extra weight is sparse. That is intended. A gate that fires on half the batch is mostly reweighting noise; a gate that fires on one in eight completed mistakes is closer to “teach the near misses.”

The wall-clock cost of the verifier is negligible next to vLLM generation. End-to-end training time stays at

about 20 minutes, the same as OPSD.

5 Analysis and Limitations

Why HMMT moves and AIME does not. AIME24/25 were already the successes of local OPSD. Adding a sparse outcome weight cannot be expected to push a saturated contest as much as an unsaturated one. HMMT25 is harder for this 1.7B checkpoint (base 23.1), and local OPSD did not help it. Reweighting completed mistakes is a plausible way to spend the same 100 steps on traces that actually reached an answer and got it wrong—precisely the traces for which a privileged teacher has something to correct. We do not have a causal token-level proof of that story; it is consistent with the numbers and with the training statistics, nothing more.

Single-seed evaluation. The OPSD README states that Avg@12 from one seed will move. A 0.8 point AIME24 change is inside that band; a 5.0 point HMMT change, with matching Pass and majority-vote lifts, is less so. A multi-seed rerun would be the right next measurement and is not in this budget.

What we did not do. We did not mix GRPO’s group advantage into the KL, did not train a process reward model, and did not change the evaluation protocol to flatter the gated model. We did not report paper 4-GPU OPSD as “our baseline.” Those choices keep the comparison honest and the method small.

Limitations. Results are for Qwen3-1.7B, LoRA, and 100 steps. The verifier only sees a boxed span; a wrong but well-reasoned trace with no box is treated like a truncation. The discrete R is not a credit assignment over tokens. Scaling the same gate to 8B, to thinking-on student rollouts, or to a full GRPO token budget is open.

6 Conclusion

OPSD already gives a dense, on-policy, single-model training signal. It does not look at whether the student was right. We add that look as a sequence weight, and we refuse to spend the extra weight on truncated text. Under a matched local budget the change does not raise AIME, and we report that. It does raise HMMT25 by five Avg@12 points, which is the contest where re-produced OPSD had failed. The intervention is a few

dozen lines on top of the public trainer, $\lambda=0$ restores the original algorithm, and the baselines are the ones we actually ran.

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